

## Subnetting Mastery & Mock Exam

1. Given the network 192.168.10.0/24, divide it into 4 equal subnets and write down the network address, first usable IP, last usable IP, and broadcast address for each subnet.

->To divide **192.168.10.0/24** into **4 equal subnets**, borrow 2 host bits:

**/24 → /26**

- Subnet mask: **255.255.255.192**
- Each subnet has **64 total addresses**
- Each subnet has **62 usable addresses**

```
{
  "learning_viz": {
    "type_id": "IPV4_SUBNETTING_CIDR",
    "initial_values": {
      "address_octet_1": 192,
      "address_octet_2": 168,
      "address_octet_3": 10,
      "address_octet_4": 0,
      "prefix_length": 26
    }
  }
}
```

| Subnet | Network Address       | First Usable IP | Last Usable IP | Broadcast Address     |
|--------|-----------------------|-----------------|----------------|-----------------------|
| 1      | <b>192.168.10.0</b>   | 192.168.10.1    | 192.168.10.62  | <b>192.168.10.63</b>  |
| 2      | <b>192.168.10.64</b>  | 192.168.10.65   | 192.168.10.126 | <b>192.168.10.127</b> |
| 3      | <b>192.168.10.128</b> | 192.168.10.129  | 192.168.10.190 | <b>192.168.10.191</b> |
| 4      | <b>192.168.10.192</b> | 192.168.10.193  | 192.168.10.254 | <b>192.168.10.255</b> |

### Final Answer

The four equal subnets are:

1. 192.168.10.0/26
2. 192.168.10.64/26
3. 192.168.10.128/26
4. 192.168.10.192/26

Each provides **62 usable IP addresses**.

2. You are setting up a Zomato-style food delivery office with 3 departments: Customer Support (50 devices), Delivery Management (30 devices), and Marketing (20 devices). Design a subnetting plan using the 10.0.0.0/24 network so that each department has its own subnet, maximizing available addresses and minimizing waste. List the subnet mask and IP range for each department.

->For **10.0.0.0/24**, use **VLSM** so each department gets the smallest suitable subnet.

### Subnet requirements

- **Customer Support: 50 devices** → /26 gives **62 usable IPs**
- **Delivery Management: 30 devices** → /27 gives **30 usable IPs**
- **Marketing: 20 devices** → /27 gives **30 usable IPs**

### Subnetting Plan

| Department          | CIDR                | Subnet Mask     | Network Address | Usable IP Range               | Broadcast  |
|---------------------|---------------------|-----------------|-----------------|-------------------------------|------------|
| Customer Support    | <b>10.0.0.0/26</b>  | 255.255.255.192 | 10.0.0.0        | <b>10.0.0.1 – 10.0.0.62</b>   | 10.0.0.63  |
| Delivery Management | <b>10.0.0.64/27</b> | 255.255.255.224 | 10.0.0.64       | <b>10.0.0.65 – 10.0.0.94</b>  | 10.0.0.95  |
| Marketing           | <b>10.0.0.96/27</b> | 255.255.255.224 | 10.0.0.96       | <b>10.0.0.97 – 10.0.0.126</b> | 10.0.0.127 |

### Why this is efficient

- Customer Support gets **62 usable addresses** for 50 devices.
- Delivery Management gets **30 usable addresses** for exactly 30 devices.
- Marketing gets **30 usable addresses** for 20 devices.
- The remaining **10.0.0.128–10.0.0.255** is available for future departments or expansion.

This uses **VLSM**, so the larger department gets a larger subnet while smaller departments don't consume unnecessary address space.

3. .Troubleshoot the following scenario: Two teams in your Flipkart warehouse are accidentally assigned overlapping subnets (Team A: 172.16.10.0/25, Team B: 172.16.10.64/26). Identify the overlapping IP range and suggest a corrected subnet plan so both teams have unique IP ranges.<br><br><em><strong>Hint:</strong> Draw the address ranges for both subnets to spot the overlap.</em>

->The problem is that **Team B's subnet is completely inside Team A's subnet**, so their IP ranges overlap.

### 1. Identify the ranges

#### Team A: 172.16.10.0/25

- Network: 172.16.10.0
- Usable: 172.16.10.1 – 172.16.10.126
- Broadcast: 172.16.10.127

#### Team B: 172.16.10.64/26

- Network: 172.16.10.64
- Usable: 172.16.10.65 – 172.16.10.126
- Broadcast: 172.16.10.127

### 2. Overlapping range

The overlapping **usable IP range** is:

**172.16.10.65 – 172.16.10.126**

Both teams are also claiming the same network/broadcast boundary (172.16.10.64–127).

### 3. Corrected subnet plan

If Team A needs **up to 126 usable addresses** and Team B needs **up to 62 usable addresses**, we can move Team B to the next non-overlapping /26 subnet:

| Team   | CIDR             | Usable IP Range                      | Broadcast     |
|--------|------------------|--------------------------------------|---------------|
| Team A | 172.16.10.0/25   | <b>172.16.10.1 – 172.16.10.126</b>   | 172.16.10.127 |
| Team B | 172.16.10.128/26 | <b>172.16.10.129 – 172.16.10.190</b> | 172.16.10.191 |

Now the ranges are completely separate.

**Important:** 172.16.10.128/26 is a valid /26 subnet because /26 boundaries occur at .0, .64, .128, and .192.

4. You are managing a growing IPL Fantasy League app backend and notice that your 192.168.1.0/26 subnet is running out of IPs. Propose a solution to handle address exhaustion so new servers can be added without disrupting existing users.  
  
**Hint:** Consider subnet resizing or migrating to a larger subnet.

->The current network is **192.168.1.0/26**, which provides only **62 usable IP addresses**. If the IPL Fantasy League backend is growing, a larger subnet is needed.

**Recommended solution: Expand to /25**

Move the backend to:

**192.168.1.0/25**

- Subnet mask: **255.255.255.128**
- Network address: **192.168.1.0**
- Usable range: **192.168.1.1 – 192.168.1.126**
- Broadcast: **192.168.1.127**
- Usable addresses: **126**

This doubles the usable address capacity from **62 to 126**.

**How to avoid disrupting existing users**

A safe migration could be:

1. **Plan the new /25 subnet** and reserve the additional addresses.
2. Keep existing servers using their current IPs (192.168.1.1–192.168.1.62) where possible.
3. Add new servers using the newly available range (192.168.1.64–192.168.1.126).
4. Update DHCP, routing, firewall rules, and server configurations as required.
5. Test connectivity before moving production services.

**Alternative for larger future growth**

If the application is expected to grow significantly, consider a **/24**:

## 192.168.1.0/24 → 254 usable addresses

This provides much more room for servers and infrastructure, but it requires ensuring that the larger network does not overlap with another existing network.

**Best approach:** A /25 is a reasonable immediate expansion because it provides **126 usable addresses** while retaining the existing 192.168.1.0–62 addresses.

5. After completing your subnetting speed round and mock exam, review your answers and identify 3 types of mistakes you made (e.g., miscalculating subnet masks, overlapping ranges, wrong broadcast address). For each, write how you would avoid the mistake next time.

### ->Subnetting Mistake Review

After reviewing my subnetting practice, I identified these **three common mistakes** and how I would avoid them:

| Mistake                               | What went wrong                                                                                              | How I would avoid it next time                                                                                                                         |
|---------------------------------------|--------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>1. Miscalculating subnet masks</b> | I sometimes confused CIDR values such as /26 and /27.                                                        | Memorize the common masks: /24 = 255.255.255.0, /25 = 255.255.255.128, /26 = 255.255.255.192, /27 = 255.255.255.224, /28 = 255.255.255.240.            |
| <b>2. Wrong broadcast address</b>     | I sometimes selected the last usable IP instead of the final address of the subnet as the broadcast address. | First calculate the <b>block size</b> , then identify the next subnet boundary. The address immediately before that boundary is the broadcast address. |
| <b>3. Overlapping subnet ranges</b>   | When assigning multiple subnets, I sometimes started the next subnet before the previous subnet had ended.   | Write down each subnet's <b>network → usable range → broadcast</b> before assigning the next one. Check that no ranges overlap.                        |

### Key rule I will follow

**CIDR → subnet mask → block size → network address → broadcast address → usable range**

Following this order will help me solve subnetting questions more accurately and avoid repeating these mistakes.

